

Dataset
(data that is used to
perform analysis or
intelligence extraction)

Population
(Universal Set)

eg XYZ company

- 4 branches

DEL, BOM, CHN, BLR

- each branch has 100 emps

If I consider a typical emp
dataset, the given dataset can
be called as population if it contains
all 400 records

RESULT OF ANALYSIS IS ALWAYS
ACCURATE.

Sample
(Subset of the population that represents
the entire population)

X

sample_emp.csv →

10 records

9 - DEL

1 - BOM

the given dataset can be called
as sample if the dataset
contain BALANCED record
of each branch.

8 records

4 3 2 DEL

1 1 2 BOM

1 2 2 CHN

2 2 2 BLR

RESULT OF ANALYSIS IS ALWAYS
APPROXIMATE

(margin of error will be present
and needs to be considered
during reporting)

Deal with Samples

Result/Outcomes of stat operation is always APPROXIMATE ← STATISTICS → a way to get information out of the data (sample)

~ 70%

DESCRIPTIVE STATISTICS (understanding and preparing data)

Data Exploration Phase

EDA → Exploratory Data Analysis

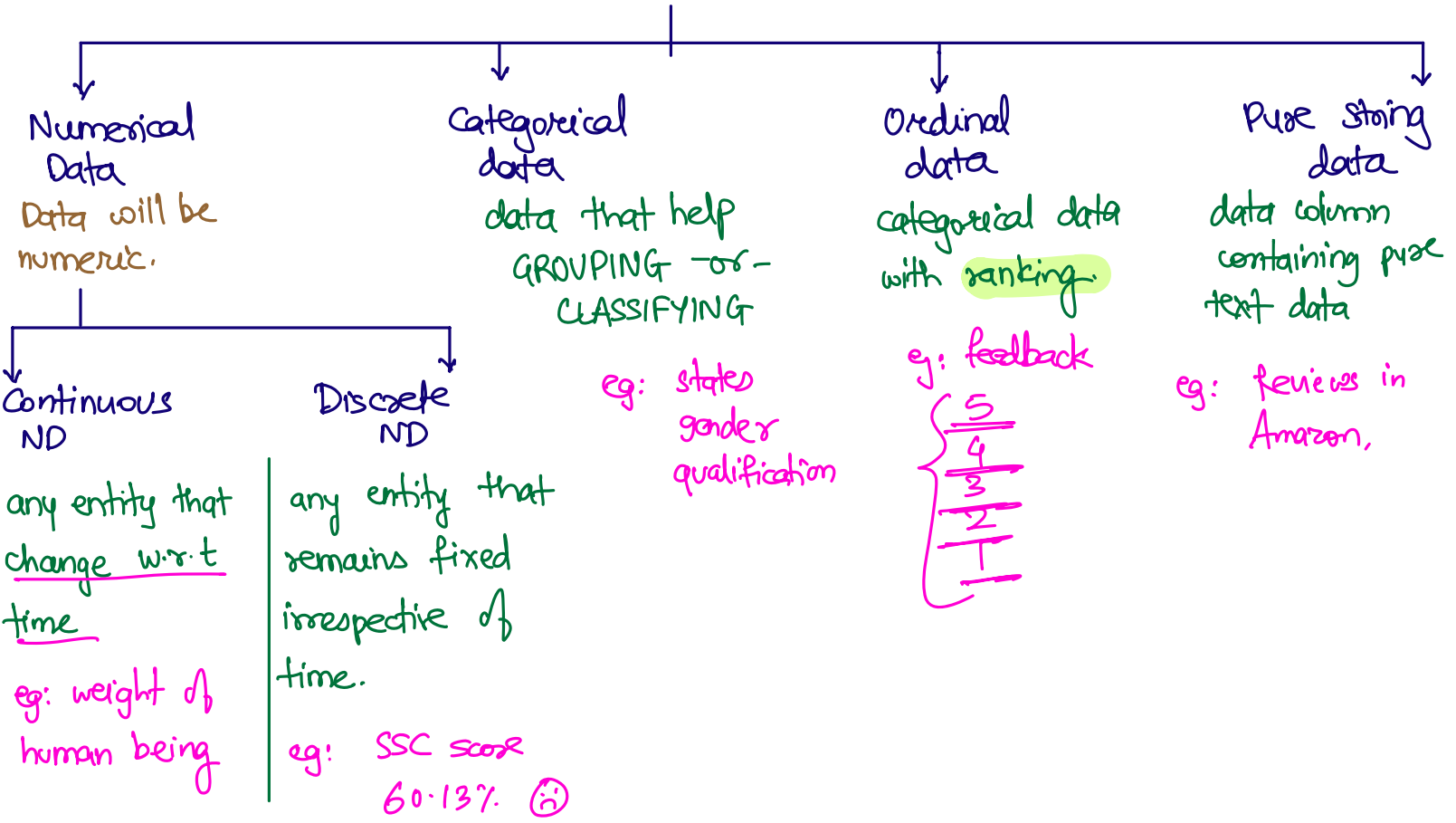
- help you ORGANISE and SUMMARIZE the data.
- identify the quality of the data w.r.t given problem statement
- To make your data COMPATIBLE for INTELLIGENCE extraction.
(Data preparation phase)

~ 30%

(extracting value out of the data)
INFERENTIAL STATISTICS
Data Modelling Phase

- Create Predictive Models
- technique of measuring the reliability of CONCLUSIONS or PREDICTIONS of population based on given sample

Types of data columns in a typical dataset



Guideline (PN's guidelines)

Historical data

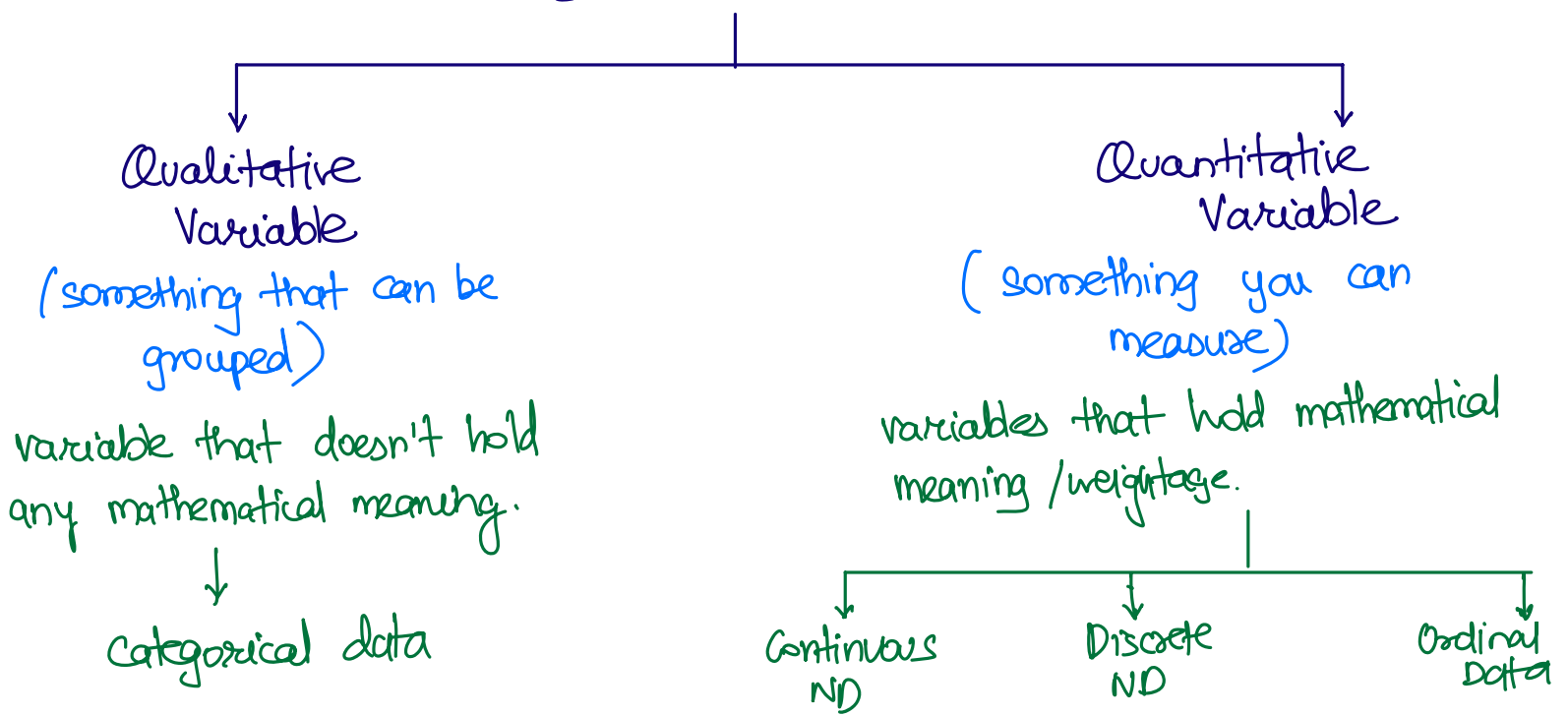
if range is defined by the domain expert in ND:

DISCRETE ND

else:

CONTINUOUS ND.

Types of Variables (columns)



Team

Data Analyst
(analysis using STATS Rules.)

Business Analyst
(analysis based on Domain rules and stat rules)

Data Artist
(responsible for building reports, PPTS, dashboards etc)

* Domain Experts
(responsible to map domain knowledge in stats)

↓

(Jack of all trades)
Data Scientist (GOD)
(leader)

Creates and Orchestrates
Data & AI Team

talks / negotiates / present
findings to client
stakeholder.

Descriptive Statistics

Data exploration phase - or - EDA (Exploratory Data Analysis)

Descriptive stats is the first step to deal with SAMPLE. Typical operations that a data analyst, data scientist, business analyst prefer includes:

- ① Check the type of DISTRIBUTION for each numerical column.
- ② Identify and deal with INAPPROPRIATE data.
- ③ Identify and deal with OUTLIERS.
- ④ Identify and deal with CATEGORICAL data
- ⑤ Identify and deal with MISSING VALUES
- ⑥ Identify and deal with ORDINAL data
- ⑦ Discover any

ASSOCIATION
RELATIONSHIP
PATTERN

between two or more variables (columns) in the given dataset.

Dataset Procurement Stage

Internal
Dataset

data is generated, managed
and maintained by same
orgn=

External
Dataset.

data that is sourced from
third party